

F. Additional Quantitative Results

F.1. Transformation Generalization

F.1.1. Detailed analysis.

For the instance shown in Table 1, from in-distribution to composition, POOD, and OOD, the exact match decreases from 1 to 0.01, 0, and 0, and the edit distance increases from 0 to 0.13, 0.17 when tested on data with transformation $f_1 \circ f_1$. Apart from ID, LLMs cannot produce a correct full chain in most cases, while they can produce correct CoT reasoning when exposed to some composition and POOD conditions by accident. As shown in Table 2, from $f_1 \circ f_2$ to $f_2 \circ f_2$, the LLMs can correctly answer 0.1% of questions. A close examination reveals that it is a coincidence, e.g., the query element is A, N, A, N, which happened to produce the same result for the two operations detailed in the Appendix G. When further analysis is performed by breaking the full chain into reasoning steps and answers, we observe strong consistency between the reasoning steps and answers. For example, under the composition generalization setting, the reasoning steps are entirely correct on test data distribution $f_1 \circ f_1$ and $f_2 \circ f_2$, but with wrong answers. Probe these insistent cases in Appendix G, we can find that when a novel transformation (say $f_1 \circ f_1$) is present, LLMs try to generalize the reasoning paths based on the most similar ones (i.e., $f_1 \circ f_2$) seen during training, which leads to correct reasoning paths, yet incorrect answer, which echo the example in the introduction. Similarly, generalization from $f_1 \circ f_2$ to $f_2 \circ f_1$ or vice versa allows LLMs to produce correct answers that are attributed to the commutative property between the two orthogonal transformations with unfaithful reasoning paths. Collectively, the above results indicate that the CoT reasoning fails to generalize to novel transformations, not even to novel composition transforms. Rather than demonstrating a true understanding of text, CoT reasoning under task transformations appears to reflect a replication of patterns learned during training.

F.1.2. Introducing another transformation.

Our main conclusion is that CoT reasoning cannot generalize to genuinely novel transformations, including unseen compositions, even when the underlying primitives are well learned. Counterexamples based on commutativity have already substantiated this claim (Table 2). We now extend the analysis by introducing a non-commutative transformation f_3 and evaluating generalization behaviors that cannot be trivially explained by commutative equivalence. The results are summarized in Table 4.

When models are trained on mixtures of transformations involving f_2 and f_3 and evaluated on unseen compositions (e.g., $f_2 \circ f_3, f_3 \circ f_2, f_3 \circ f_3 \rightarrow f_2 \circ f_2$), performance remains extremely poor across all metrics. Exact match accuracy for the full chain stays below 10%, and both edit distance and BLEU scores indicate substantial divergence from the correct reasoning traces and final answers. Notably, even when the reasoning component occasionally achieves high exact match (e.g., 100% reasoning accuracy in $f_2 \circ f_3, f_3 \circ f_2, f_2 \circ f_2 \rightarrow f_3 \circ f_3$), the corresponding full-chain and answer-level accuracy collapse, revealing a clear disconnect between locally plausible reasoning steps and globally correct execution.

More strikingly, in strictly non-commutative transfer settings such as $f_2 \circ f_3 \rightarrow f_3 \circ f_2$ and its reverse, the model fails completely: exact match drops to 0 across reasoning, answer, and full chain, while edit distance sharply increases and BLEU scores approach zero. Unlike earlier commutativity-induced cases—where incorrect reasoning paths could still yield correct answers—these failures demonstrate that once superficial equivalences are removed, CoT reasoning no longer exhibits any meaningful transfer. This provides strong evidence that prior apparent generalization was not driven by learning transformation semantics, but rather by exploiting distributional artifacts such as commutativity.

Overall, Table 4 reinforces our central claim: CoT reasoning does not support systematic generalization to novel transformations. Instead, its success hinges on structural overlap and distributional shortcuts present in the training data. When these shortcuts are eliminated via non-commutative transformations, both reasoning traces and answers degrade simultaneously, exposing the brittleness of CoT reasoning under genuine task-level distribution shifts. Illustrative examples of f_3 are provided in Appendix B for completeness

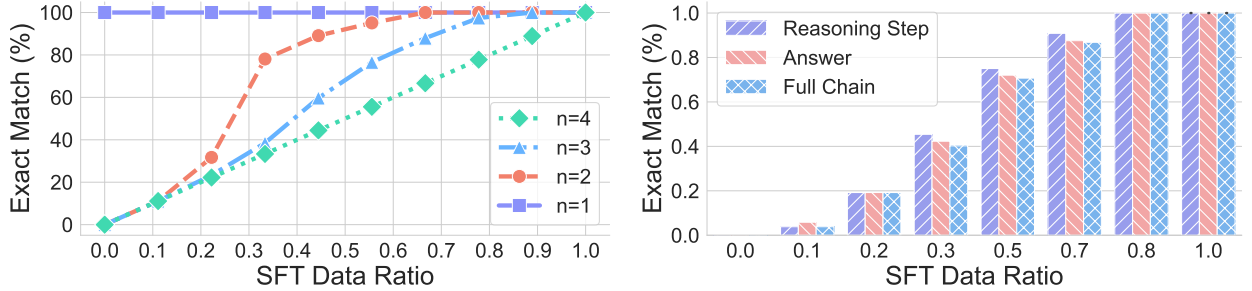
F.2. Element Generalization

Element generalization is another critical factor to consider when LLMs try to generalize to new tasks.

Experiment settings. Similar to transformation generalization, we fix other factors and consider three progressive distribution shifts for elements: ID, CMP, and OOD, as shown in Figure 2. It is noted that in composition, we test if CoT reasoning can be generalized to novel combinations when seeing all the basic atoms in the elements, e.g., $(A, B, C, D) \rightarrow (B, C, D, A)$. Based on the atom order in combination (can be measured by edit distance n), the CMP can be further developed. While for OOD, atoms that constitute the elements are totally unseen during the training.



Figure 11 | Element generalization results on various scenarios and relations.



(a) Performance on unseen element via SFT in various CMP (b) Evaluation of CoT reasoning in SFT. scenarios.

Figure 12 | SFT performances for element generalization. SFT helps to generalize to novel elements.

Findings. Similar to transformation generalization, the performances degrade sharply when facing the distribution shift consistently across all transformations, as shown in Figure 11. From ID to CMP and OOD, the exact match decreases from 1.0 to 0 and 0, for all cases. Most strikingly, the BLEU score is 0 when transferred to f_1 and f_2 transformations. A failure case in Appendix G shows that the models cannot respond to any words when novel elements are present. We further explore when CoT reasoning can generalize to novel elements by conducting SFT. The results are summarized in Figure 12. We evaluate the performance under three exact matches for the full chain under three scenarios, CMP based on the edit distance n . The result is similar to SFT on transformation. The performance increases rapidly when presented with similar (a small n) examples in the training data. Interestingly, the exact match rate for CoT reasoning aligns with the lower bound of performance when $n = 3$, which might suggest the generalization of CoT reasoning on novel elements is very limited, even SFT on the downstream task. When we further

analyze the exact match of reasoning, answer, and token during the training for $n = 3$, as summarized in Figure 12b. We find that there is a mismatch of accuracy between the answer and the reasoning step during the training process, which somehow might provide an explanation regarding why CoT reasoning is inconsistent in some cases.

Table 5 | Evaluation on text length generalization.

Text Length	Exact Match (%)			Edit Distance			BLEU Score		
	Reasoning	Answer	Full Chain	Full Chain	Reasoning	Answer	Full Chain	Reasoning	Answer
2	0.00	0.00	0.00	0.3772	0.4969	0.5000	0.4214	0.1186	0.0000
3	0.00	0.00	0.00	0.2221	0.3203	0.2540	0.5471	0.1519	0.0000
4	100.00	100.00	100.00	0.0000	0.0000	0.0000	1.0000	1.0000	1.0000
5	0.00	0.00	0.00	0.1818	0.2667	0.2000	0.6220	0.1958	0.2688
6	0.00	0.00	0.00	0.3294	0.4816	0.3337	0.4763	0.1174	0.2077

F.3. Text Length Generalization

Text length generalization evaluates how CoT performance varies when the input text length (i.e., the element length l) differs from training examples. Considering the way LLMs process long text, this aspect is crucial because real-world problems often involve varying degrees of complexity that manifest as differences in problem statement length, context size, or information density.

Experiment settings. We pre-train LLMs on the dataset with text length merely on $l = 4$ while fixing other factors and evaluate the performance on a variety of lengths. We consider three different padding strategies during the pre-training: (i) None: LLMs do not use any padding. (ii) Padding: We pad LLM to the max length of the context window. (iii) Group: We group the text and truncate it into segments with a maximum length.

Findings. As illustrated in the Table 5, the CoT reasoning failed to directly generate two test cases even though those lengths present a mild distribution shift. Further, the performance declines as the length discrepancy increases shown in Figure 13. For instance, from data with $l = 4$ to those with $l = 3$ or $l = 5$, the BLEU score decreases from 1 to 0.55 and 0.62. Examples in Appendix G indicate that LLMs attempt to produce CoT reasoning with the same length as the training data by adding or removing tokens in the reasoning chains. The efficacy of CoT reasoning length generalization deteriorates as the discrepancy increases. Moreover, we consider using a different padding strategy to decrease the divergence between the training data and test cases. We found that padding to the max length doesn't contribute to length generalization. However, the performance increases when we replace the padding with text by using the group strategy, which indicates its effectiveness.

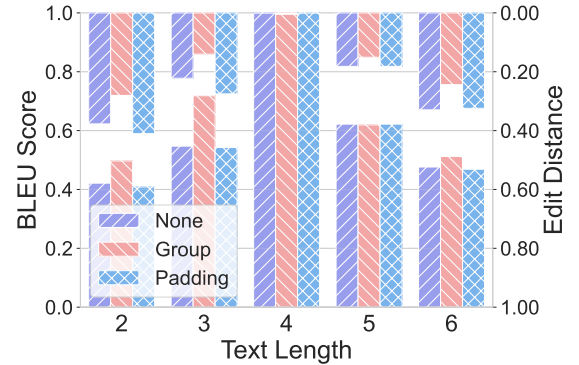


Figure 13 | Performance of text length generalization across various padding strategies. Group strategies contribute to length generalization.

F.4. Temperature and Model Size

Temperature and model size generalization explores how variations in sampling temperature and model capacity can influence the stability and robustness of CoT reasoning. For the sake of rigorous evaluation, we further investigate whether different choices of temperatures and model sizes may